



Factory-to-Customer Shipping Route Efficiency Dashboard

Nassau Candy Distributor

Dataset Shape: (10194, 26)

Columns

- ▼ [
- 0 : "Row ID"

1 : "Order ID"

2 : "Order Date"

3 : "Ship Date"

4 : "Ship Mode"

5 : "Customer ID"

6 : "Country/Region"

7 : "City"

8 : "State/Province"

9 : "Postal Code"

10 : "Division"

11 : "Region"

12 : "Product ID"

13 : "Product Name"

14 : "Sales"

15 : "Units"

16 : "Gross Profit"

17 : "Cost"

18 : "Original_Lead_Time"

19 : "Lead Time"

20 : "Factory"

21 : "Route_State"

22 : "Route_Region"

23 : "Shipping_Category"

24 : "Recommended_Shipping"

25 : "Delayed"

]

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Custc
0	1	US-2021-103800-CHO-MIL-31000	2024-01-03 00:00:00	2026-06-30 00:00:00	Standard Class	
1	2	US-2021-112326-CHO-TRI-54000	2024-01-04 00:00:00	2026-07-01 00:00:00	Standard Class	
2	3	US-2021-112326-CHO-NUT-13000	2024-01-04 00:00:00	2026-07-01 00:00:00	Standard Class	
3	4	US-2021-112326-CHO-SCR-58000	2024-01-04 00:00:00	2026-07-01 00:00:00	Standard Class	
4	5	US-2021-141817-CHO-TRI-54000	2024-01-05 00:00:00	2026-07-05 00:00:00	Standard Class	

 Dashboard Overview

 Total Orders

10,194

 Average Lead Time

1320.84 ...

 Average Cost

\$4.74

 Average Gross Profit

\$9.17

 Total Routes

196

 States Covered

59

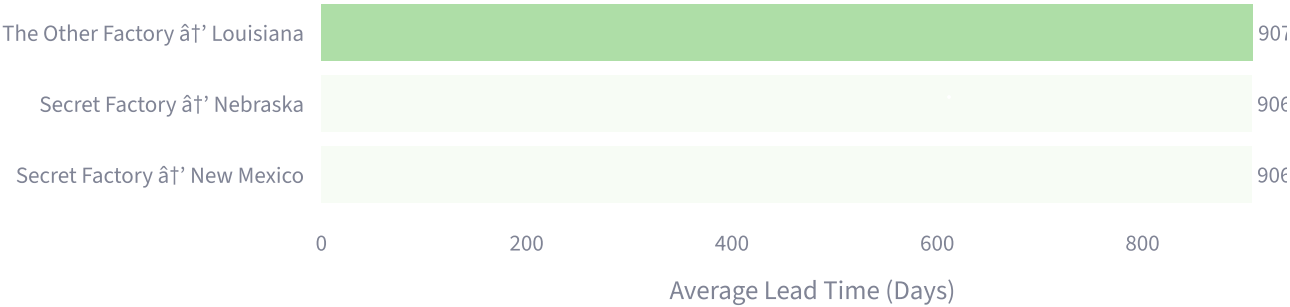
 Delay Percentage

100.0%

 Route Performance Analysis

Top 10 Fastest Factory-to-Customer Routes





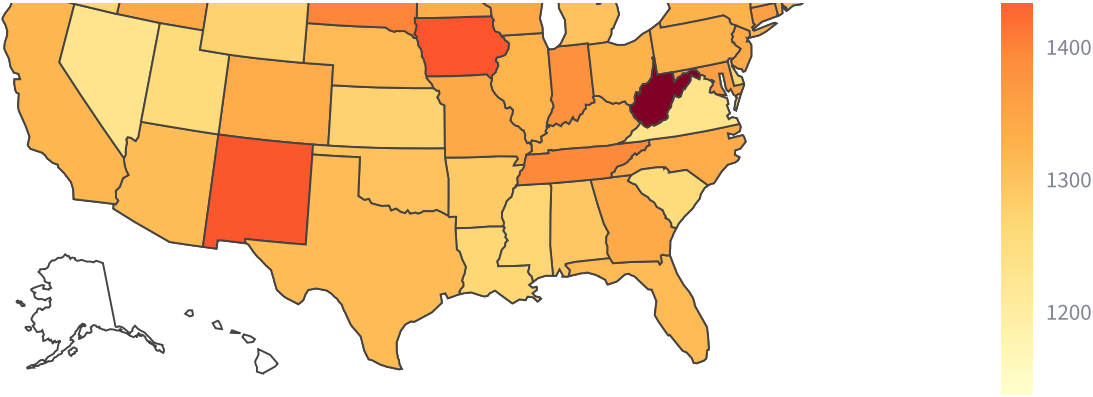
 Route Performance Summary

Route_State	Total_Shipments	Avg_Lead_Time	Lead_Time_Variability
Secret Factory to New Mexico	2	906	2.83
Secret Factory to Nebraska	1	906	0
The Other Factory to Louisiana	1	907	0
The Other Factory to Connecticut	2	907.5	2.12
Secret Factory to Mississippi	1	908	0
Wicked Choccy's to Maine	2	908	0
Secret Factory to Louisiana	2	908.5	0.71
Secret Factory to Delaware	1	909	0
Secret Factory to South Carolina	1	909	0
Secret Factory to Minnesota	1	909	0

 Geographic Shipping Performance

Average Shipping Lead Time Across U.S. States





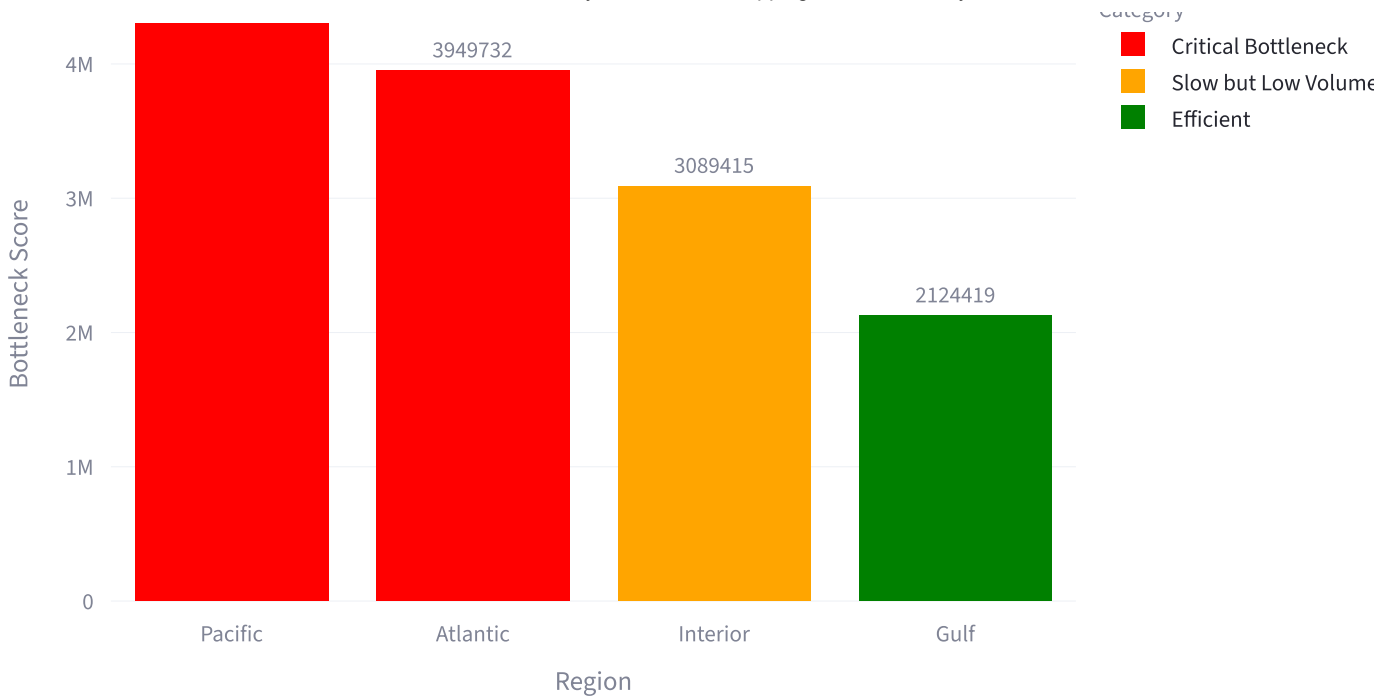
State Shipping Performance

State/Province	Total_Shipments	Avg_Lead_Time	State_Code
West Virginia	4	1638	WV
North Dakota	7	1637.86	ND
Saskatchewan	2	1457	None
Manitoba	12	1455.33	None
Iowa	30	1443.9	IA
New Mexico	37	1441.84	NM
Vermont	11	1438.91	VT
Prince Edward Island	10	1420.3	None
South Dakota	12	1395.92	SD
Tennessee	183	1391.49	TN



Regional Bottleneck Analysis

Regional Bottleneck Score



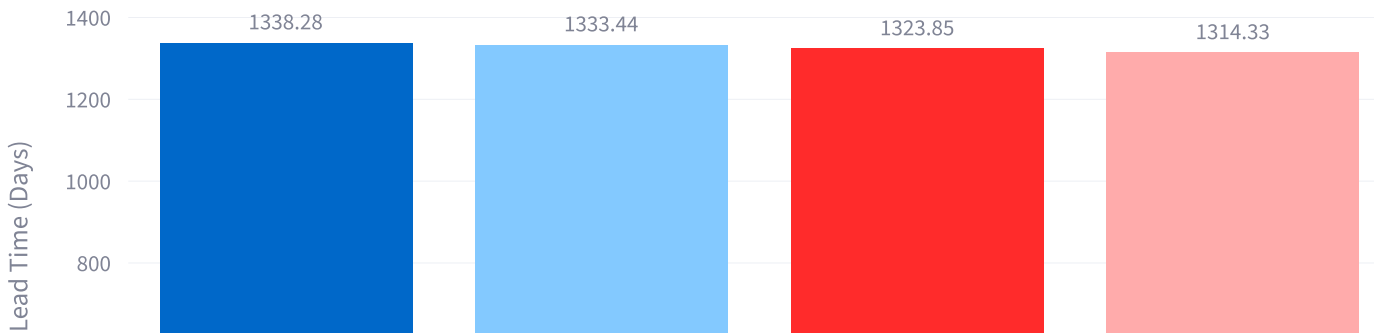
Regional Performance Summary

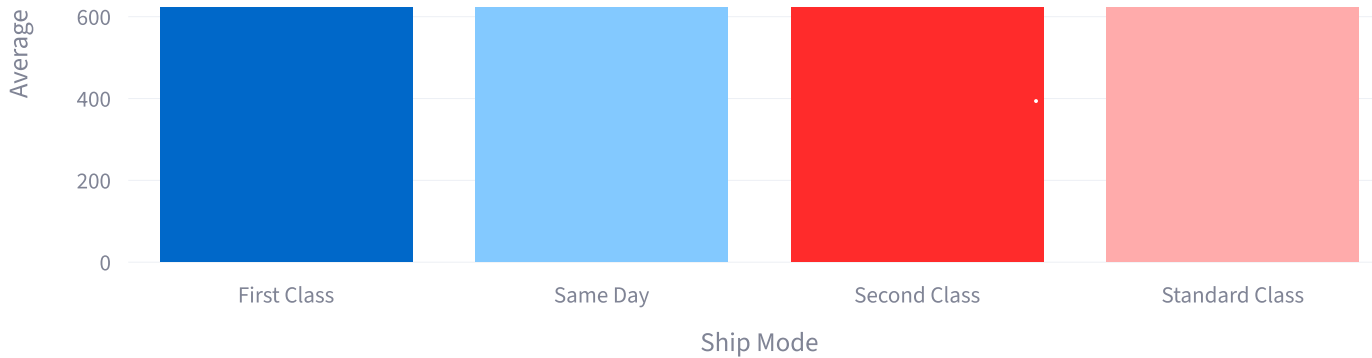
Region	Total_Shipments	Avg_Lead_Time	Lead_Time_Variability	Bottleneck_Score	Category
Pacific	3253	1322.19	266.47	4301084	Critical Bottleneck
Atlantic	2986	1322.75	256.54	3949732	Critical Bottleneck
Interior	2335	1323.09	262.69	3089415	Slow but Low Volume
Gulf	1620	1311.37	264.73	2124419	Efficient



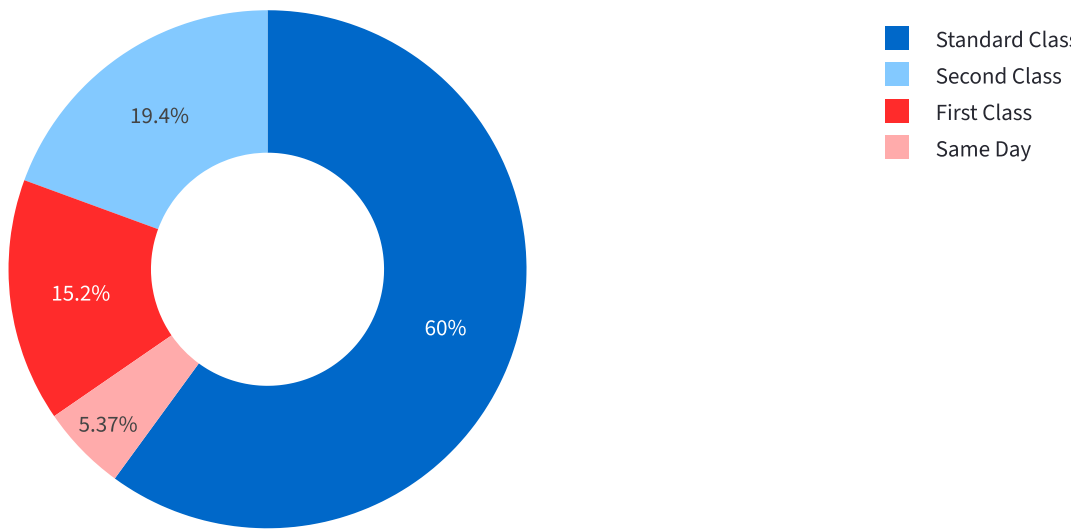
Ship Mode Performance Comparison

Average Shipping Lead Time by Ship Mode





Shipment Distribution by Ship Mode



 Ship Mode Performance Summary

Ship Mode	Total_Shipments	Avg_Lead_Time	Avg_Cost	Avg_Gross_Profit	Lead_Time_Variability	Efficiency
Standard Class	6120	1314.33	4.75	9.22	262.4	
Second Class	1979	1323.85	4.83	9.25	261.81	
Same Day	547	1333.44	4.41	8.59	253.81	
First Class	1548	1338.28	4.72	9.05	265.63	

Best Performing Ship Mode: Standard Class

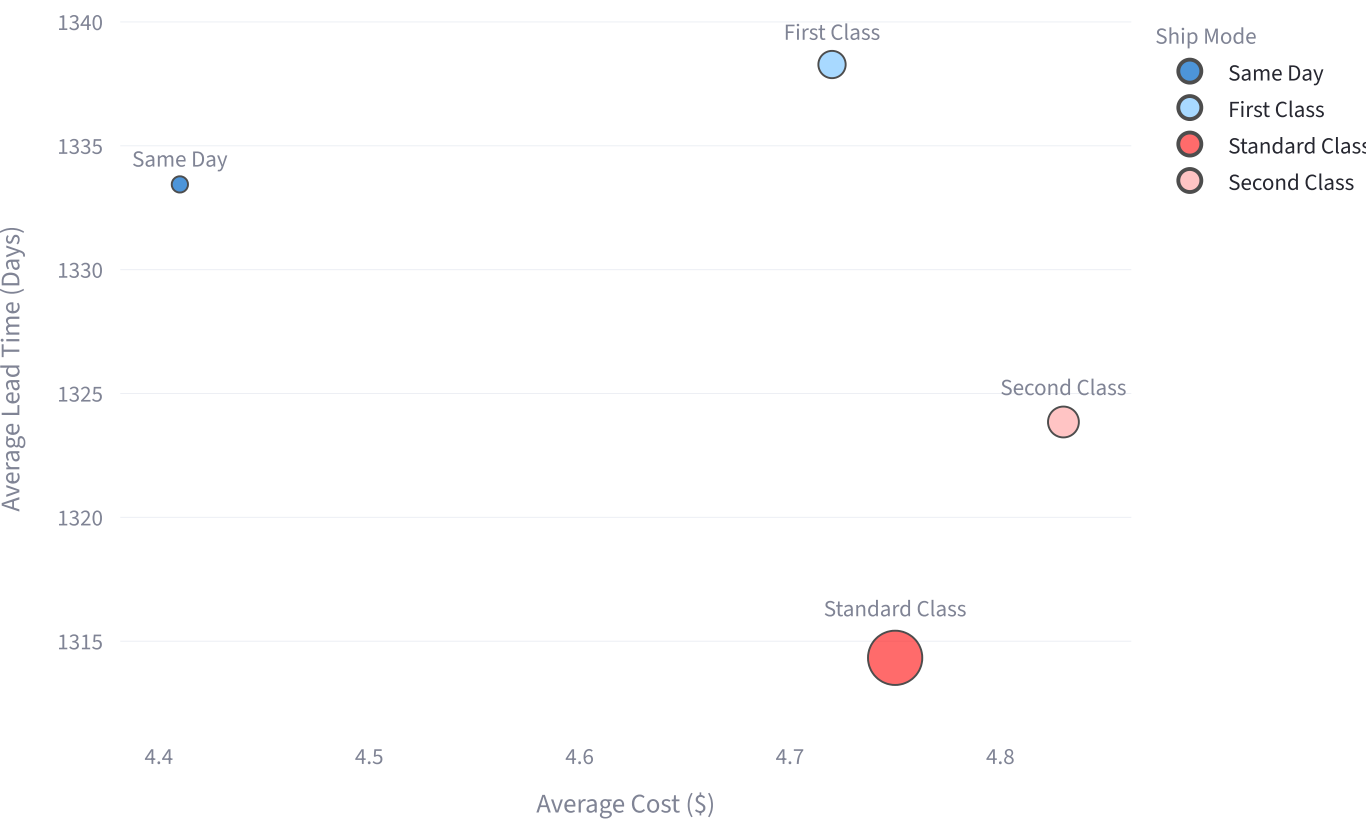
- Average Lead Time : 1314.33 Days
- Average Cost : \$4.75

• Total Shipments : 6120

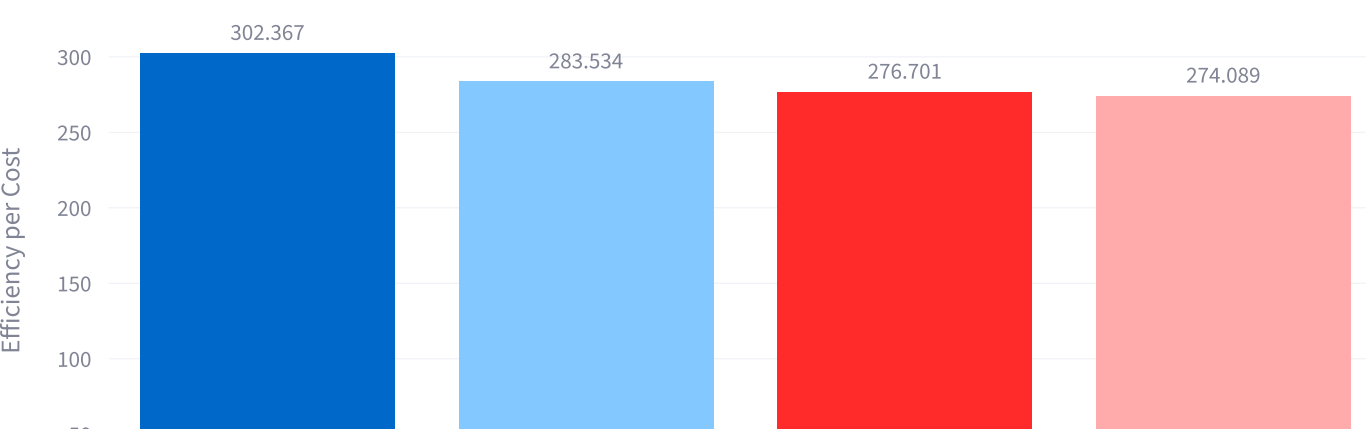


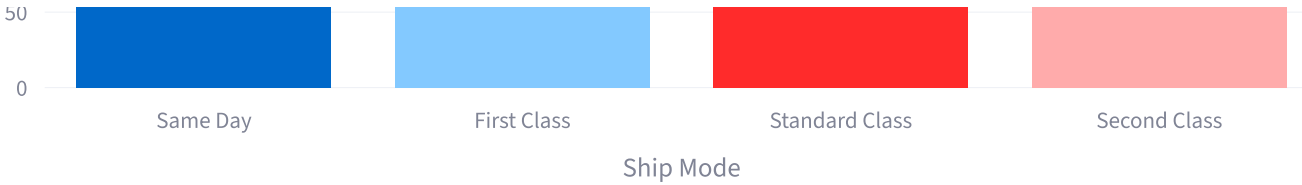
Efficiency vs Cost Analysis

Efficiency vs Cost by Ship Mode



Efficiency per Cost Ranking





Efficiency vs Cost Summary

Ship Mode	Avg_Lead_Time	Avg_Cost	Total_Shipments	Efficiency_per_Cost
Same Day	1333.44	4.41	547	302.367
First Class	1338.28	4.72	1548	283.534
Standard Class	1314.33	4.75	6120	276.701
Second Class	1323.85	4.83	1979	274.089



Best Cost-Efficient Shipping Mode

Ship Mode: Same Day

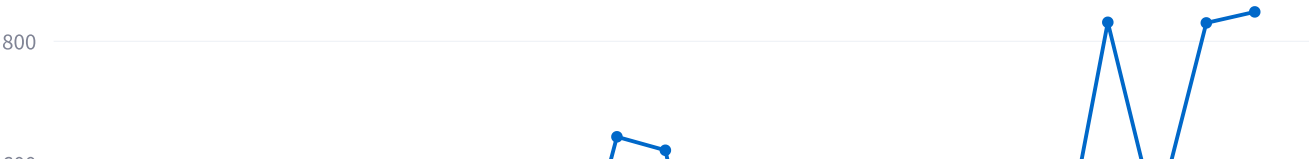
- Efficiency per Cost : 302.367
- Average Lead Time : 1333.44 Days
- Average Cost : \$4.41
- Total Shipments : 547

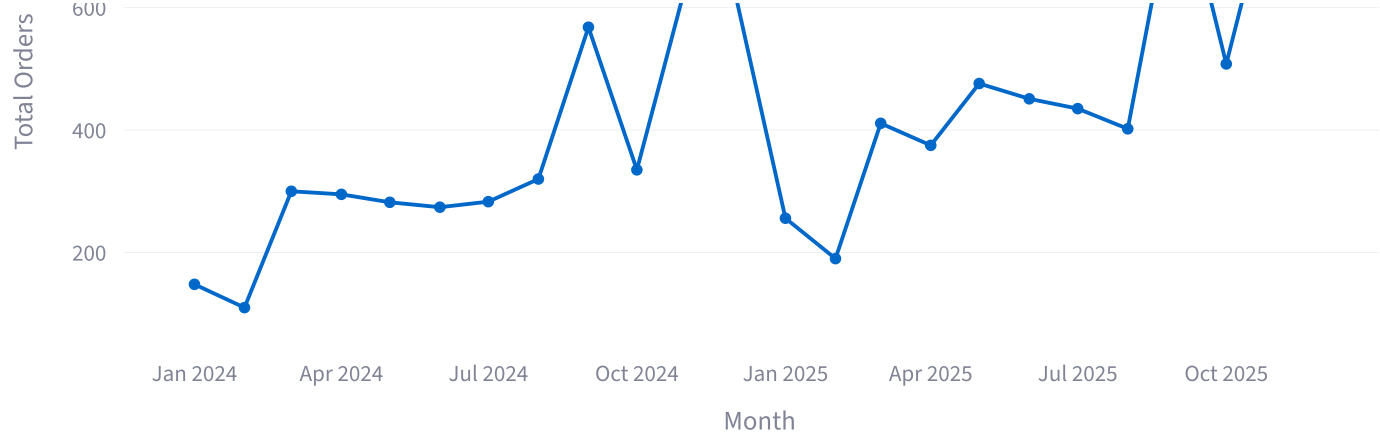
This shipping mode provides the highest operational efficiency relative to its average transportation cost.



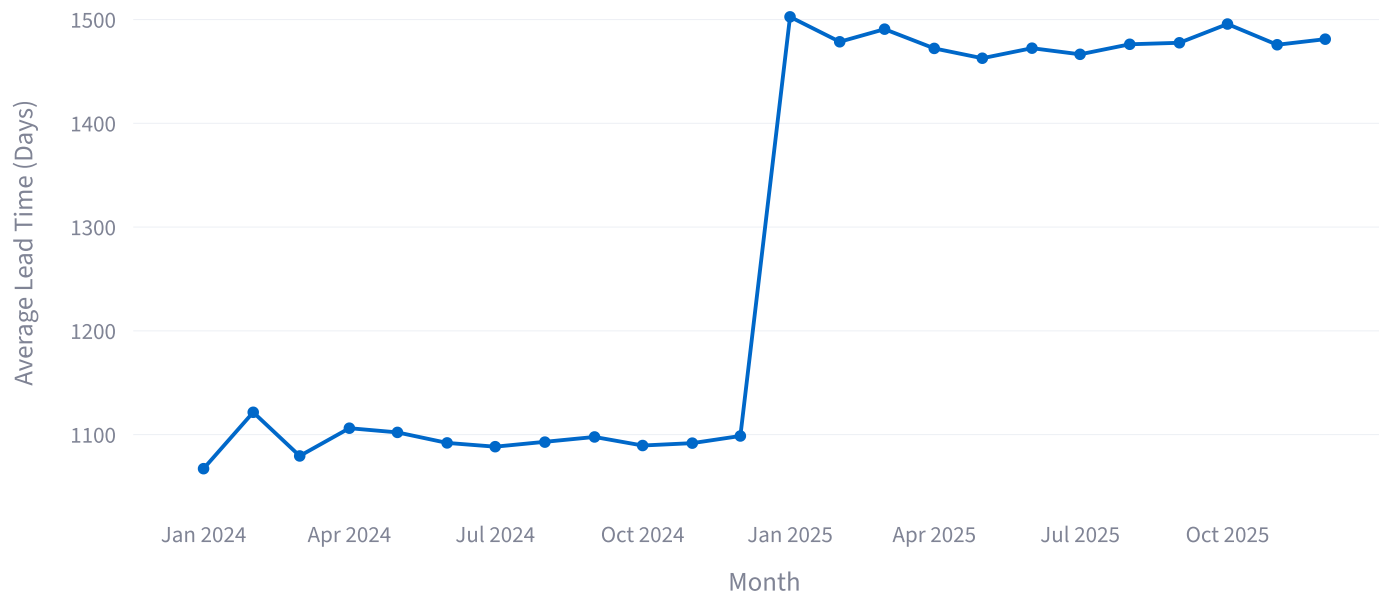
Shipping Timeline Analysis

Monthly Order Volume

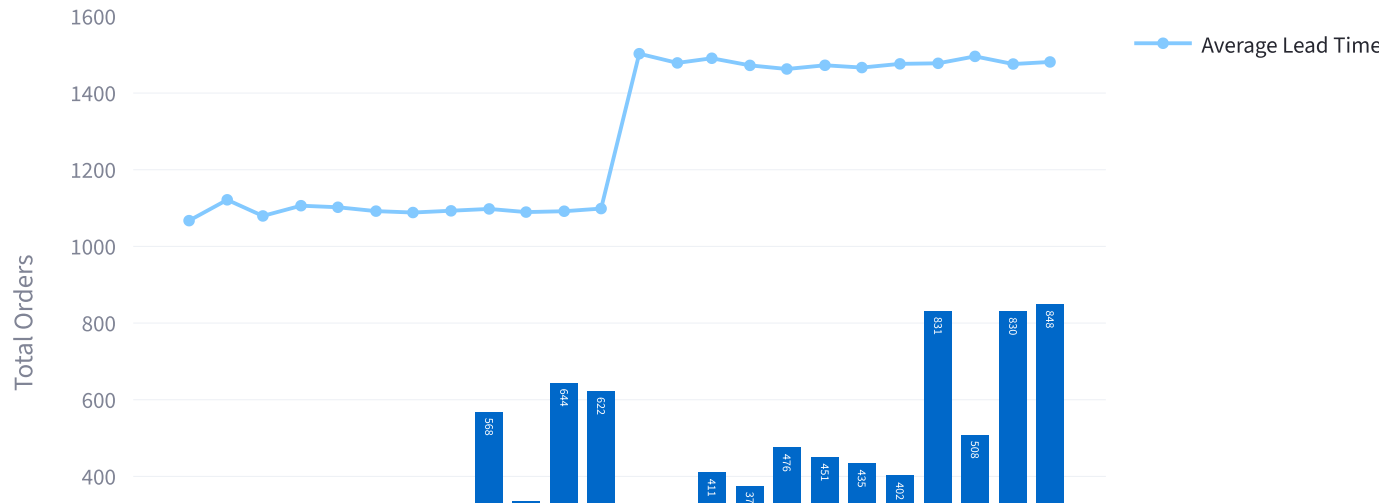


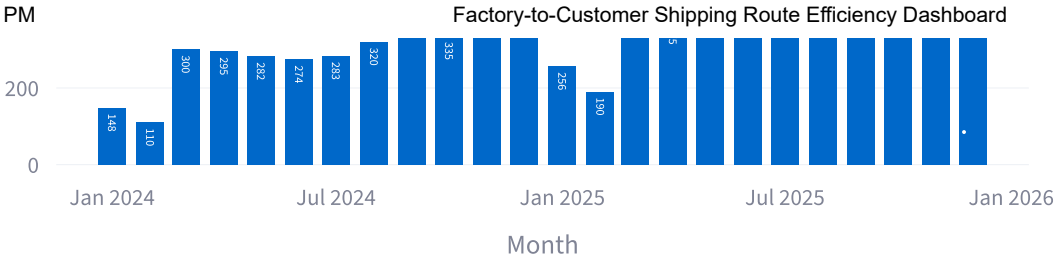


Monthly Average Shipping Lead Time



Orders and Average Lead Time by Month





Monthly Shipping Performance

Month	Total_Orders	Avg_Lead_Time	Avg_Cost	Avg_Gross_Profit
2024-01	148	1067.16	4.92	9.23
2024-02	110	1121.5	4.28	8.26
2024-03	300	1079.42	4.89	8.9
2024-04	295	1106.17	4.66	8.86
2024-05	282	1102.12	4.74	9.27
2024-06	274	1091.99	4.34	8.51
2024-07	283	1088.35	4.59	9.06
2024-08	320	1092.75	4.79	9.21
2024-09	568	1097.73	4.76	9.17
2024-10	335	1089.47	5.06	9.31



Timeline Insights

- Highest shipment volume occurred in **2025-12** with **848 orders**.
- Fastest average delivery performance was achieved in **2024-01** with an average lead time of **1067.16 days**.



Route Drill-Down Analysis

Select a Factory-to-Customer route to explore detailed shipment performance.

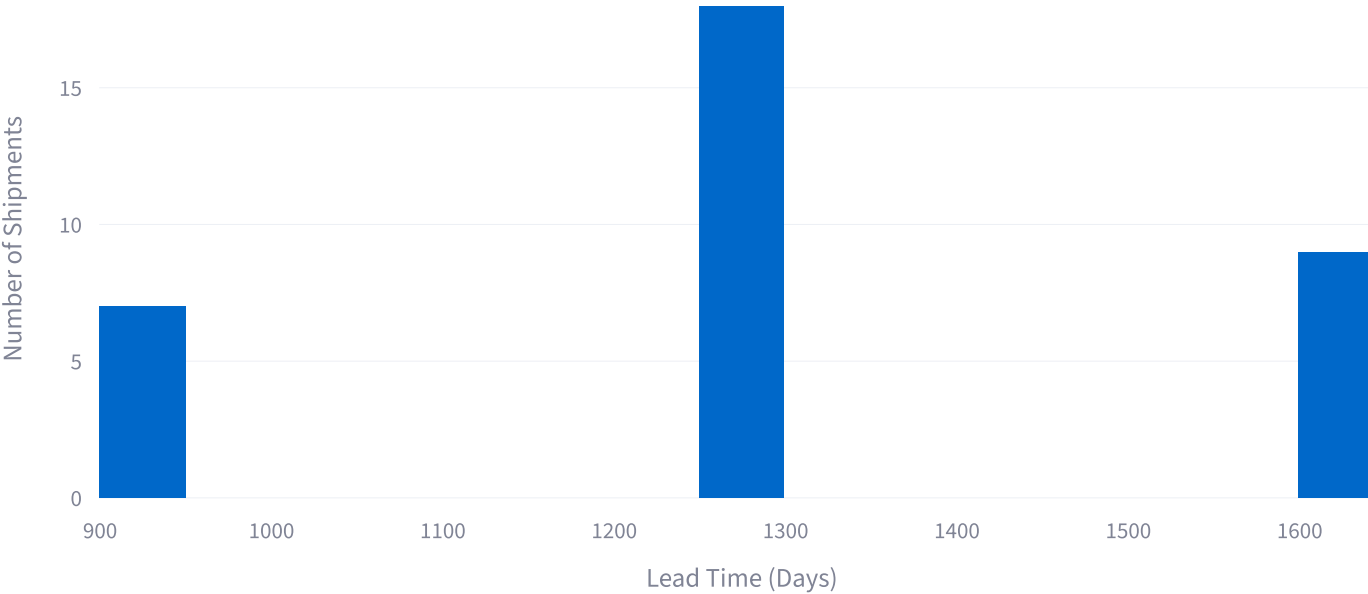
Select Route

Lot's O' Nuts â†’ Alabama

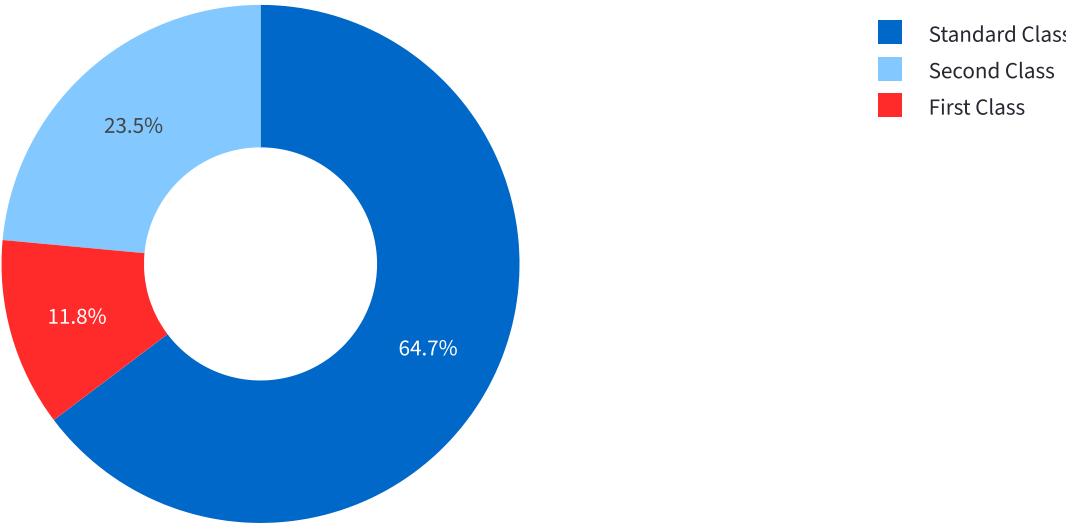
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Shipments	Avg Lead Time	Avg Cost	Avg Profit	Lead Time Std
34	1295....	\$4.37	\$9.62	253.07

Lead Time Distribution
Lot's O' Nuts â†’ Alabama



Ship Mode Distribution





Shipment Details

Order ID	Order Date	Ship Date	Ship Mode	State/Province	Cost
US-2022-134257-CHO-NUT-13000	2024-03-16 00:00:00	2027-09-09 00:00:00	Second Class	Alabama	2
US-2022-134257-CHO-FUD-51000	2024-03-16 00:00:00	2027-09-09 00:00:00	Second Class	Alabama	6
US-2022-121552-CHO-NUT-13000	2024-03-22 00:00:00	2027-09-17 00:00:00	Standard Class	Alabama	4
US-2021-118997-CHO-FUD-51000	2024-04-08 00:00:00	2026-10-03 00:00:00	Standard Class	Alabama	9.6
US-2021-143840-CHO-FUD-51000	2024-05-22 00:00:00	2026-11-19 00:00:00	Standard Class	Alabama	2.4
US-2021-143840-CHO-NUT-13000	2024-05-22 00:00:00	2026-11-19 00:00:00	Standard Class	Alabama	1
US-2022-104941-CHO-SCR-58000	2024-06-13 00:00:00	2027-12-10 00:00:00	Standard Class	Alabama	3.3
US-2022-104941-CHO-SCR-58000	2024-06-13 00:00:00	2027-12-10 00:00:00	Standard Class	Alabama	3.3
US-2022-104941-CHO-SCR-58000	2024-06-13 00:00:00	2027-12-10 00:00:00	Standard Class	Alabama	5.5
US-2022-154291-CHO-FUD-51000	2024-07-25 00:00:00	2028-01-19 00:00:00	Standard Class	Alabama	10.8

Route Summary

Selected Route: Lot's O' Nuts â†’ Alabama

- **Total Shipments:** 34
- **Average Lead Time:** 1295.21 Days
- **Average Transportation Cost:** \$4.37
- **Average Gross Profit:** \$9.62
- **Lead Time Variability:** 253.07 Days

This section enables detailed operational monitoring of individual factory-to-customer routes and helps identify high-performing as well as underperforming logistics routes.



Download Filtered Data

Download the filtered shipment data for further analysis in Excel, Power BI, or other reporting tools.



Download Filtered Dataset (CSV)



Filtered Dataset Summary

Metric	Value
Total Records	10194
Total Routes	196
States Covered	59
Average Lead Time (Days)	1320.84
Average Cost (\$)	4.74
Average Gross Profit (\$)	9.17

The downloaded CSV contains only the records matching the selected dashboard filters.

Dashboard Notes

- Order-to-Shipment Duration is calculated directly from the Order Date and Ship Date available in the dataset.
- Dashboard filters update all KPIs, charts and tables dynamically.
- Developed using Python, Streamlit and Plotly.

Project: Factory-to-Customer Shipping Route Efficiency Analysis for Nassau Candy Distributor